

CALIMAN ADVISORY

TECHNICAL HANDOFF DOCUMENT

Executive Financial Dashboard

Power BI semantic model

Architecture, conventions, known issues and maintenance guide for whoever maintains or extends this project next.

VERSION 1.0 • JULY 2026

The dataset behind this project is synthetic and describes no real entity.

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1 Purpose and audience

This document hands the Executive Financial Dashboard over to whoever maintains or extends it next: a BI developer, an FP&A analyst comfortable with DAX, or a controller who owns the reporting output. It records how the semantic model is built, the conventions that keep the financial statements balanced, the defects that are known and deliberately visible, and the checks to run before publishing any change.

The project is a Power BI Desktop file in PBIP format (Executive Financial Statements.pbip). Report pages follow a dark executive theme: black background, white typography, red accent. The companion deliverables, a management report and an executive review deck, were generated from this model via DAX and ship alongside this document.

2 Model architecture

The model is a star schema. Two fact tables carry GL postings, one for actuals and one for budget, at account and posting-date grain. Conformed dimensions for accounts and dates filter both facts. Three structure tables drive the layout of the financial statement matrices; they define row order, indentation and hierarchy for the income statement, balance sheet and cash flow visuals. All amounts are in euros.

2.1 Tables

Table	Role	Notes
GL_Actual	Fact	General ledger actuals, posted through April 2026. One resident measure; all others live in M_Measures.
GL_Budget	Fact	Budget postings through December 2026, positive magnitudes by convention.
CAPEX	Fact	Capital expenditure detail, joined to its own local date table.
Opening_Balances	Fact	Year-end balance sheet snapshots per year. Account 330100 is the current-year result. The accounting equation balances per year.
Dim_Accounts	Dimension	Chart of accounts with FSGroup mapping and a precomputed sign column used by the Actual and Budget base measures.
Dim_Date	Dimension	Conformed calendar for both GL facts.
Dim_FSGroup	Dimension	Financial statement grouping above the account level.
INCOME_Structure / BS_Structure / CF_Structure	Structure	Row templates for the three statement matrices. Report-value measures read their row context.
M_Measures	Measure home	Single-column container holding the measure library (321 measures).

2.2 Relationships

All relationships are many-to-one, single direction, filtering from dimension to fact. There are no bidirectional relationships in the model, which keeps filter propagation predictable and is intentional.

From (many)	To (one)	Status
GL_Actual[Account]	Dim_Accounts[Account]	Active
GL_Budget[Account]	Dim_Accounts[Account]	Active
GL_Actual[PostingDate]	Dim_Date[Date]	Active
GL_Budget[BudgetDate]	Dim_Date[Date]	Active
Dim_Accounts[FSGroup]	Dim_FSGroup[FSGroup]	Active
CAPEX[CapexDate]	Auto local date table	Active
Opening_Balances[OB_Date]	Auto local date table	Active

Two auto-generated local date tables exist because auto date/time is still enabled in the file. They add size without adding value, since Dim_Date already covers time intelligence. Recommended cleanup:

disable auto date/time, rebind CAPEX and Opening_Balances date filtering to Dim_Date, and confirm no visual relied on the built-in hierarchies.

3 Measure library

All measures live in M_Measures, organized by display folder. Naming is prefix-based: IS_ for income statement, BS_ for balance sheet, CF_ for cash flow, PL_ for the actual/budget/variance column set, ES_ for the executive summary page. Before writing a new measure, search the library; most column variants (actual, budget, variance, variance percent, monthly) already exist.

Folder	Contents
Income Statement - P&L	PL_* actual, budget, variance and margin measures for every statement line.
Income Statement - Monthly / Monthly Columns	Per-month column measures for the twelve-month matrix; header rows return 0 with an empty display format.
Income Statement - Margins / Time Intelligence / Forecast	Margin percentages, YTD and MTD variants, run-rate forecast measures.
Balance Sheet - Statement Columns / _Base / Ratios / Working Capital / YoY / Targets	Year columns (BS_2023 through BS_2026), cumulative GL bases, liquidity and leverage ratios, threshold targets for conditional formatting.
Cash Flow (+ Ratios)	IAS 7 indirect-method components, opening and closing cash, free cash flow and quality-of-earnings ratios.
Formatting folders	Font and background color measures consumed by conditional formatting; they return hex strings, not numbers.
JSON Measures	Measures that emit JSON or full HTML for the HTML Content visual on the executive summary page (ES_HTML and companions).

3.1 Conventions that must not be broken

- Sign handling is centralized. The base [Actual] and [Budget] measures apply a sign precomputed on Dim_Accounts through a single vectorized SUMX over the fact table. Do not reintroduce per-account SUMX over the dimension with context transition; that pattern previously pushed the statement matrices past the Power BI Service per-visual memory limit.
- Statement visuals read structure tables. The matrices bind to INCOME_Structure, BS_Structure and CF_Structure rows, and the report-value measures (FS Report Value, BS_ReportValue, CF_ReportValue) resolve what to show from row context. Changing a statement layout means editing the structure table, not the visual.
- BS_CheckBalance must return zero for every period. It asserts assets equal liabilities plus equity. Run it after any change that touches balance sheet measures or account mapping.

- Equity is derived from the accounting identity (assets minus liabilities), not rolled forward independently. This was a deliberate 7 July 2026 change; it guarantees the statement balances and surfaces data problems in equity movement instead of hiding them.
- Liabilities present positive through ABS() at presentation level only; cash flow delta measures keep raw signs for their own arithmetic. Keep presentation sign logic out of calculation measures.
- Assets and liabilities display positive by convention, with one deliberate exception: the negative inventory anomaly is shown at true sign (see section 5).

4 Data sources and refresh

Source data is a synthetic general ledger maintained in an Excel workbook loaded through Power Query. Actual postings currently exist through April 2026 and budget postings through December 2026. Refresh is manual from Power BI Desktop. There is no gateway, no incremental refresh and no RLS; none are needed at portfolio scale. If the project is ever pointed at a live ledger, revisit all three.

5 Known issues, open by design

These defects were found while validating the statements and are documented rather than patched cosmetically. Fixing the data is the right correction; masking the symptoms in DAX is not.

Item	Detail	Suggested owner
Negative inventory	Inventory is €(7.9)M at 30 April 2026. Physical stock cannot be negative; the likely cause is COGS relief posted without matching receipts. The per-component ABS() that used to mask it was removed on 7 July 2026.	Accounting / data generation
Cash cross-tie	Balance sheet cash (€1.6M) and cash flow closing cash (€5.4M) disagree by €3.8M, and the cash flow opening anchor is negative. The two figures come from different calculation paths. Each statement is internally consistent; the cross-statement tie fails.	BI developer
DSO / DPO	BS_DSO returns over 1,800 days because the standalone AR and inventory measures ignore the date filter in some contexts and return cumulative all-time values. The composed statement measures (CF_AsOf_*) are correct; the standalone ones are not. Exclude DSO and DPO from reporting until rebased.	BI developer
CF_CashConversionRatio	Returns 2.08x for YTD April 2026, while operating cash flow over net income computes to 1.28x. The measure logic needs review before the ratio is quoted anywhere.	BI developer
IS_Forecast	Under a whole-year filter the measure returns the YTD actual rather than an annualized run rate. Verify the intended grain before using it on report pages.	BI developer
Budget baseline	The 2026 budget implies a full-year EBITDA loss against actuals running at a 34% margin. Variances against it are not decision-useful. A reforecast is the fix; this is a planning issue, not a model bug.	Controller / FP&A

6 Validation checklist before publishing changes

- BS_CheckBalance = 0 for every year from 2023 through the current year.
- P&L internal ties: gross profit = revenue minus COGS; EBITDA = gross profit minus OPEX; EBIT = EBITDA minus D&A; EBT = EBIT plus financial result; net income = EBT minus taxes.
- Cash flow internal tie: opening cash plus net movement = closing cash.
- OPEX account detail sums to the OPEX total measure.
- Statement matrices render within the Service per-visual memory limit (test in the Service, not only Desktop).
- No new bidirectional relationships and no duplicated measures.

The DAX used to validate this handoff is reproducible: filter Dim_Date to the target period and evaluate the PL_*, BS_* and CF_* measures listed in sections 3 and 5 with SUMMARIZECOLUMNS or CALCULATETABLE(ROW(...)). All figures in the July 2026 management report and executive deck came from those queries, unadjusted.

7 Extension ideas

- Replace the auto date tables as described in section 2.2.
- Add a reforecast fact table alongside GL_Budget so variance reporting can switch baselines mid-year.
- Rebase DSO and DPO on the CF_AsOf_* measures and reintroduce them to the balance sheet page.
- Add a calculation group for Actual / Budget / PY / Variance to retire some of the duplicated PL_* column measures.